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Original Research Article

Development and Optimization of Glibenclamide-Loaded Chitosan Microspheres in a Thermosensitive Nasal In-Situ Gel: In-Vitro Formulation and Characterization

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Study scope: Non-animal, non-human formulation development and in-vitro characterization

Abstract

Background: Glibenclamide is a potent sulfonylurea whose formulation performance is constrained by poor aqueous solubility and variable oral absorption. A particulate-in-gel nasal formulation was investigated as a formulation strategy to improve drug retention and provide controlled in-vitro release.

Objective: To formulate glibenclamide-loaded chitosan microspheres, optimize the formulation using a 3^2 full factorial design, and incorporate the optimized microspheres into a thermosensitive in-situ gel for physicochemical and in-vitro performance characterization.

Methods: Chitosan microspheres were prepared by a water-in-oil emulsion cross-linking process using glutaraldehyde. Chitosan concentration (X1) and glutaraldehyde volume (X2) were evaluated at three levels. Particle size (Y1), entrapment efficiency (Y2), and 8-h cumulative drug release (Y3) were used as formulation responses. The optimized formulation was characterized for particle size, polydispersity, zeta potential, release kinetics, sol-to-gel transition temperature, mucoadhesive strength, and viscosity.

Results: The reported optimized formulation F5 contained 1.5% w/v chitosan and 2.0 mL glutaraldehyde and showed a mean particle diameter of 42.35 ± 1.20 µm, entrapment efficiency of 84.62 ± 1.85%, and 8-h drug release of 81.15 ± 1.40%. The reported regression model for entrapment efficiency showed $R^2 = 0.9930$, adjusted $R^2 = 0.9813$, and adequate precision of 28.42. The in-situ gel showed a sol-to-gel transition temperature of 33.4 ± 0.3 °C and reported mucoadhesive strength of 4250 ± 112 dynes/cm². The release profile was best fitted by the Korsmeyer–Peppas model ($R^2 = 0.9945$; $n = 0.62$).

Conclusion: The reported data support successful formulation optimization and favorable in-vitro physicochemical performance of the microsphere-loaded thermosensitive gel. No animal, human-subject, animal-derived tissue, or human-derived biological material was used in this revised study scope; therefore, conclusions are limited to formulation and in-vitro characterization.

Keywords: Glibenclamide; chitosan microspheres; thermosensitive in-situ gel; Poloxamer 407; PEG 400; factorial design; drug release; Korsmeyer–Peppas kinetics.

1. Introduction

Type 2 diabetes mellitus is a chronic metabolic disorder requiring sustained pharmacological management. Glibenclamide (GLB), a second-generation sulfonylurea, acts primarily by stimulating insulin release through interaction with ATP-sensitive potassium channels in

pancreatic beta cells. Although therapeutically potent, formulation and absorption characteristics can limit the consistency of oral drug delivery.

The source manuscript describes glibenclamide as a poorly water-soluble, lipophilic drug and identifies dissolution and variable systemic exposure as important formulation challenges. Improving the microenvironment around the drug and controlling its release are therefore relevant formulation objectives.

Intranasal administration is attractive from a dosage-form development perspective because the nasal cavity provides a highly vascularized mucosal surface and can support rapid drug uptake. However, a conventional liquid preparation may be removed rapidly by mucociliary clearance. A formulation that combines particulate drug entrapment with temperature-triggered gel formation may increase residence and maintain a more controlled release environment.

Chitosan was selected as the microsphere-forming polymer because its protonatable amino groups can provide mucoadhesive interactions with negatively charged mucosal surfaces. Cross-linking with glutaraldehyde can produce a particulate polymeric matrix capable of entrapping drug. Poloxamer 407 (Pluronic F-127) was selected as the thermoresponsive gel-forming polymer, with PEG 400 included as a formulation component.

The present revised manuscript deliberately focuses on formulation development and in-vitro characterization. Animal pharmacokinetic, pharmacodynamic, histopathological, and animal-derived ex-vivo tissue experiments from the earlier source version are excluded from the study scope rather than replaced by invented experiments.

2. Materials and Methods

2.1 Study design and scope

This was a formulation-development and in-vitro characterization study. The experimental scope comprised preparation of glibenclamide-loaded chitosan microspheres, factorial optimization, physicochemical characterization, incorporation into a thermosensitive in-situ gel, and in-vitro drug-release and rheological evaluation. No animal study, human study, animal-derived tissue study, or human-derived biological material was included in the revised scope.

2.2 Materials

The source manuscript reported glibenclamide (USP, >99.2% purity), low-molecular-weight chitosan (degree of deacetylation >85%; molecular weight approximately 150 kDa), Pluronic F-127 (Poloxamer 407), polyethylene glycol 400 (PEG 400), and glutaraldehyde solution (25% v/v). Light liquid paraffin, Span 80, and glacial acetic acid were also reported. The source

identified commercial suppliers including Sun Pharmaceutical Industries Ltd., Sigma-Aldrich, and Merck India Ltd.

2.3 Preparation of glibenclamide-loaded chitosan microspheres

A water-in-oil emulsion cross-linking procedure was reported. Chitosan at 1.0–2.0% w/v was dissolved in 1.5% v/v aqueous acetic acid. Glibenclamide (100 mg) was dissolved in 2 mL methanol and mixed with the polymeric phase under probe sonication for 2 min at 40 kHz. The aqueous phase was added dropwise to 50 mL light liquid paraffin containing 1.0% w/v Span 80 and stirred at 1800 rpm for 30 min to form the emulsion.

Glutaraldehyde solution at 1.0–3.0 mL was added dropwise, followed by cross-linking at 40 °C for 3.5 h. Microspheres were isolated by centrifugation at 4500 rpm for 15 min, washed three times with petroleum ether to remove paraffin, washed with deionized water, and vacuum-dried at 40 °C for 24 h.

2.4 Factorial design and mathematical optimization

A 3^2 full factorial design was reported to evaluate the effect of two formulation variables at three levels. X1 represented chitosan concentration and X2 represented glutaraldehyde volume. The measured responses were particle size (Y1), entrapment efficiency (Y2), and 8-h cumulative drug release (Y3).

The factor levels used in the factorial design are summarized in Table 1.

Table 1. Factorial design factors and levels

Factor	−1 level	0 level	+1 level
X1: Chitosan (% w/v)	1.0	1.5	2.0
X2: Glutaraldehyde (mL)	1.0	2.0	3.0

The second-order polynomial used for the response-surface analysis was reported as: $Y = B0 + B1X1 + B2X2 + B12X1X2 + B11X1^2 + B22X2^2$.

Reported regression equations were: (1) $Y1 = 42.61 + 8.92X1 + 3.81X2 + 1.25X1X2 + 1.15X1^2 - 0.42X2^2$; (2) $Y2 = 83.94 + 10.15X1 + 4.82X2 - 0.95X1X2 - 3.12X1^2 - 1.85X2^2$; and (3) $Y3 = 80.22 - 11.45X1 - 8.92X2 - 1.10X1X2 - 2.85X1^2 + 0.15X2^2$.

The source manuscript reported ANOVA, coefficient of determination (R^2), adjusted R^2, F-values, p-values, and adequate precision as statistical criteria. The original source did not

specify a statistical software name/version; this information has not been invented in the present revision.

2.5 Experimental formulation matrix

Table 2. Experimental formulation matrix and reported responses

Run	X1/X2	Chitosan (% w/v)	GA (mL)	Particle size (µm)	Entrapment (%)	Q8 release (%)
F1	−1/−1	1.0	1.0	32.14 ± 1.20	62.30 ± 1.40	92.40 ± 2.10
F2	−1/0	1.0	2.0	35.80 ± 1.50	68.50 ± 1.10	85.10 ± 1.90
F3	−1/+1	1.0	3.0	38.20 ± 1.10	71.20 ± 1.80	74.30 ± 2.00
F4	0/−1	1.5	1.0	39.40 ± 1.40	74.10 ± 1.30	88.20 ± 1.50
F5*	0/0	1.5	2.0	42.35 ± 1.20	84.62 ± 1.85	81.15 ± 1.40
F6	0/+1	1.5	3.0	46.10 ± 1.80	86.40 ± 1.60	68.90 ± 1.80
F7	+1/−1	2.0	1.0	48.50 ± 2.00	81.20 ± 1.70	79.40 ± 2.20
F8	+1/0	2.0	2.0	52.10 ± 1.60	88.30 ± 1.50	64.20 ± 1.60
F9	+1/+1	2.0	3.0	58.90 ± 2.10	91.10 ± 1.90	52.80 ± 1.30

*F5 was identified as the optimized formulation in the source manuscript.

2.6 Preparation of thermosensitive in-situ gel

The optimized microspheres were incorporated into a thermosensitive gel matrix based on Pluronic F-127 at 18% w/v and PEG 400 at 1.5% w/v, as reported in the source manuscript. The formulation was evaluated for sol-to-gel transition temperature, mucoadhesive strength, and viscosity under different temperature and shear-rate conditions.

2.7 Particle size, polydispersity and zeta potential

The optimized microspheres were characterized for mean particle diameter, polydispersity index (PDI), and zeta potential. The reported values were 42.35 ± 2.15 µm, PDI 0.214, and +28.4 ± 1.8 mV, respectively. The source formulation matrix reported 42.35 ± 1.20 µm for F5; this difference in reported standard deviation is retained transparently rather than silently reconciled.

2.8 In-vitro drug release and kinetic modeling

Cumulative in-vitro drug release from the optimized formulation was evaluated and fitted to commonly used mathematical models. The source manuscript reported fitting to zero-order, first-order, Higuchi, Korsmeyer–Peppas, and Hixson–Crowell models.

The reported kinetic-model fitting parameters are summarized in Table 3.

Table 3. In-vitro release kinetic model fitting

Model	R^2	Reported rate constant/parameter
Zero-order	0.9215	K = 7.42 %/h
First-order	0.9652	K = 0.211 h^{-1}
Higuchi	0.9812	K = 26.85 %/h^0.5
Korsmeyer–Peppas	0.9945	K = 23.14 %/h^n; n = 0.62
Hixson–Crowell	0.9482	K = 0.082 mg^1/3/h

The reported Korsmeyer–Peppas exponent n = 0.62 was interpreted in the source manuscript as consistent with anomalous/non-Fickian release behavior.

2.9 In-vitro gel characterization

The reported sol-to-gel transition temperature was 33.4 ± 0.3 °C and the reported mucoadhesive strength was 4250 ± 112 dynes/cm^2. Viscosity was evaluated at different temperatures and shear rates.

Table 4. Viscosity of the thermosensitive gel at different temperatures and shear rates

Temperature	State	Shear rate (s^{-1})	Viscosity (cP)
25.0 °C	Sol	10	145 ± 8
25.0 °C	Sol	100	142 ± 6
34.0 °C	Gel	10	4850 ± 120
34.0 °C	Gel	50	1920 ± 85
34.0 °C	Gel	100	1110 ± 45

2.10 Statistical analysis

Factorial responses were analyzed using a second-order polynomial response-surface approach and ANOVA. Statistical significance was interpreted using the p-values reported in the source manuscript. For the entrapment-efficiency model, the reported model F-value was 84.58 with $p < 0.0001$, $R^2 = 0.9930$, adjusted $R^2 = 0.9813$, and adequate precision = 28.42.

3. Results

3.1 Effect of formulation variables on microsphere characteristics

The nine factorial runs demonstrated a progressive increase in particle size with increasing chitosan concentration and glutaraldehyde level. Particle size ranged from 32.14 ± 1.20 µm in F1 to 58.90 ± 2.10 µm in F9. Entrapment efficiency increased across the design space from 62.30 ± 1.40% in F1 to 91.10 ± 1.90% in F9. In contrast, the 8-h cumulative release generally decreased as polymer and cross-linker levels increased, ranging from 92.40 ± 2.10% in F1 to 52.80 ± 1.30% in F9.

The complete experimental formulation matrix and reported responses are presented in Table 2.

F5, containing 1.5% w/v chitosan and 2.0 mL glutaraldehyde, was identified as the optimized formulation. It provided a balance between particle size, drug entrapment, and controlled release, with reported values of 42.35 ± 1.20 µm, 84.62 ± 1.85%, and 81.15 ± 1.40%, respectively.

3.2 ANOVA of entrapment-efficiency response

Table 5. ANOVA summary for the entrapment-efficiency response

Source	Sum of squares	df	Mean square	F-value	p-value
Model	742.18	5	148.44	84.58	<0.0001
X1 (Chitosan)	618.15	1	618.15	352.22	<0.0001
X2 (Glutaraldehyde)	139.39	1	139.39	79.42	0.0028
X1×X2	3.61	1	3.61	2.06	0.2461
$X1^2$	19.47	1	19.47	11.10	0.0447
$X2^2$	6.85	1	6.85	3.90	0.1432
Residual	5.26	3	1.75	—	—
Total	747.44	8	—	—	—

The source manuscript reported $R^2 = 0.9930$, adjusted $R^2 = 0.9813$, and adequate precision = 28.42 for the entrapment-efficiency model. Chitosan concentration and glutaraldehyde volume were reported as significant contributors to the model.

The ANOVA summary for the entrapment-efficiency response is presented in Table 5.

3.3 Optimized microsphere characterization

The optimized microspheres exhibited a mean particle diameter of approximately 42 µm, PDI 0.214, and positive zeta potential of +28.4 ± 1.8 mV. These reported characteristics indicate a defined particulate system with a positive surface charge under the reported measurement conditions.

3.4 In-vitro release profile

The optimized formulation released 81.15 ± 1.40% of the drug at 8 h according to the factorial-response table. The source manuscript further reported cumulative release values reaching approximately 95.20 ± 1.30% at 12 h in the optimized formulation. The kinetic fitting showed the highest reported R^2 for the Korsmeyer–Peppas model (0.9945), with n = 0.62.

3.5 Thermosensitive gel performance

The optimized microspheres were incorporated into an 18% w/v Pluronic F-127 and 1.5% w/v PEG 400 gel system. The reported sol-to-gel transition temperature of 33.4 ± 0.3 °C lies within the temperature range relevant to nasal administration. Viscosity increased markedly on

transition from the sol state at 25 °C to the gel state at 34 °C and decreased with increasing shear rate at 34 °C, consistent with the reported shear-dependent behavior.

The reported viscosity measurements are summarized in Table 4.

4. Discussion

The formulation data demonstrate the expected relationship between polymer/cross-linker content and microsphere structure. Increasing chitosan concentration increases the amount of polymer available to form the microsphere matrix, which is associated in the reported design space with larger particle size and greater drug entrapment. Increasing glutaraldehyde concentration likewise contributes to matrix cross-linking and is associated with higher entrapment and slower release in the reported factorial data.

The release results are consistent with a matrix-controlled system in which drug diffusion and polymeric structural relaxation can both contribute to release. The Korsmeyer–Peppas model produced the highest reported coefficient of determination, and the exponent of 0.62 was classified in the source manuscript as non-Fickian/anomalous release. This interpretation should be regarded as a mathematical description of the reported release curve rather than proof of a specific molecular release mechanism.

The thermosensitive gel provides a second formulation-control element. At room temperature the formulation remains comparatively less viscous, whereas the reported transition near 33.4 °C is accompanied by a large increase in viscosity. Such a transition can be useful for dosage-form handling and retention; however, actual nasal residence time, mucociliary clearance, permeation enhancement, and systemic bioavailability cannot be established from the present in-vitro data alone.

The positive zeta potential of the optimized microspheres is also relevant to formulation characterization because chitosan-based systems can exhibit cationic surface characteristics. Nevertheless, surface charge measurements alone do not establish mucoadhesion or enhanced epithelial permeability in vivo. The present conclusions are therefore intentionally limited to the measured formulation attributes.

5. Study limitations

The revised study intentionally excludes animal and human experimentation. Accordingly, pharmacokinetic exposure, pharmacodynamic glucose lowering, systemic bioavailability, histopathological safety, and in-vivo nasal residence cannot be concluded from this manuscript. Animal-derived ex-vivo mucosal permeation data from the earlier source version are also excluded because the present study scope is non-animal and non-human.

The source manuscript did not provide the statistical software/version, complete raw replicate-level datasets for every assay, full instrument specifications, chromatographic/bioanalytical validation details, or complete optimization desirability settings. These details should be supplied from laboratory records if a target journal requires them. No missing values have been invented in this revision.

6. Conclusion

Glibenclamide-loaded chitosan microspheres were reported to be successfully prepared and optimized using a 3^2 factorial design. The identified F5 formulation contained 1.5% w/v chitosan and 2.0 mL glutaraldehyde and showed a reported particle size of 42.35 ± 1.20 µm, entrapment efficiency of 84.62 ± 1.85%, and 8-h cumulative release of 81.15 ± 1.40%. Incorporation into an 18% w/v Pluronic F-127/1.5% w/v PEG 400 thermosensitive gel produced a reported sol-to-gel transition temperature of 33.4 ± 0.3 °C and mucoadhesive strength of 4250 ± 112 dynes/cm². Release kinetics were best described by the Korsmeyer–Peppas model. The revised manuscript supports formulation-level and in-vitro conclusions only and does not claim animal or human efficacy, safety, permeation, or bioavailability.

Ethics statement

Ethics approval: Not applicable. No animals, human participants, animal-derived tissues, or human-derived biological materials were used in the revised study scope.

Conflict of interest

The authors declare no conflict of interest.

Funding statement

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None reported.

Data availability

Formulation and characterization results reported in the study are presented within the manuscript. Additional raw data should be supplied from laboratory records where available.

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